HOCHSCHULE
RAVENSBURG-WEINGARTEN
UNIVERSITY
OF APPLIED SCIENCES

Electronic Circuit Design

Slides to the lecture

Chapter 4: Voltage References

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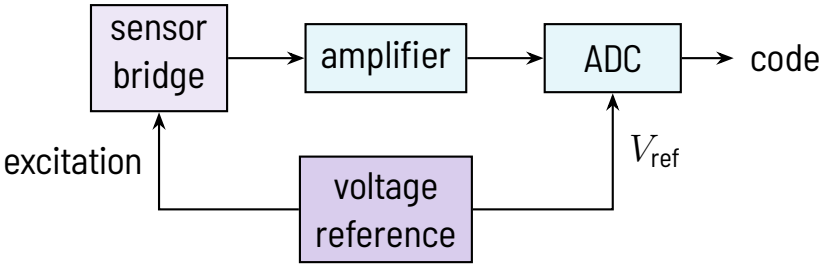
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4 Voltage References

4.1 Why Voltage References?

The reference sets the scale of the whole chain



- ADC: code $= 2^N \cdot V_{\text{in}}/V_{\text{ref}}$ (an ADC measures a **ratio**)
- bridge: output is a fraction of the excitation voltage
- an error of V_{ref} is a **gain error** of the complete chain

Regulator or reference?

	voltage regulator	voltage reference
purpose	supply power	define a voltage
initial accuracy	$1\,\%$ … $5\,\%$	$0.02\,\%$ … $1\,\%$
temperature coefficient	100 ppm/K … 300 ppm/K	1 ppm/K … 50 ppm/K
output current	mA … A	µA … 10 mA
noise	high	low, specified

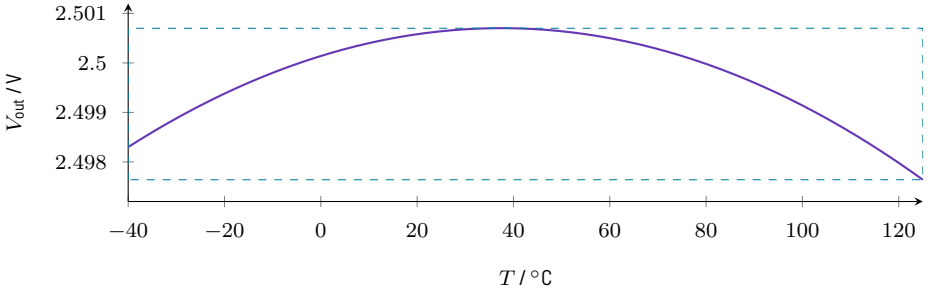
- a reference is a “voltage standard” on the board, not a power supply
- if a load needs current: buffer the reference with an op-amp

4.2 Specifications

Data sheet parameters

- **initial accuracy**: tolerance at $25\,^{\circ}\mathrm{C}$, e.g. $\pm 0.1\,\%$
- **temperature coefficient** (TC): e.g. 20 ppm/K
- **line regulation**: $\Delta V_{\text{out}}/\Delta V_{\text{in}}$, e.g. 50 µV/V
- **load regulation**: $\Delta V_{\text{out}}/\Delta I_{\text{out}}$, e.g. 100 ppm/mA
- **noise**: 0.1 Hz ... 10 Hz peak-to-peak, e.g. 10 µV pp
- **long-term drift**: e.g. 50 ppm per 1000 h
- **thermal hysteresis**: change after a temperature cycle

Temperature coefficient: the box method

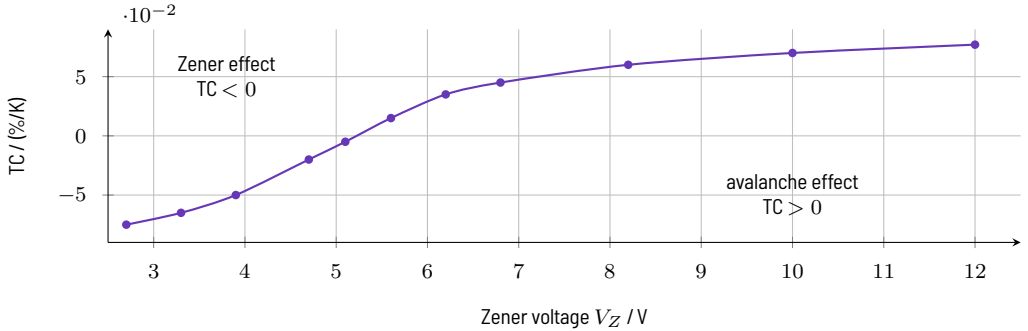


Box method

$$TC = \frac{V_{max} - V_{min}}{V_{nom}\,(T_{max} - T_{min})} \cdot 10^6 \text{ ppm}$$

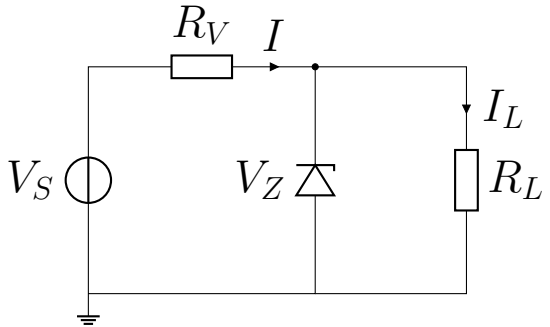
4.3 The Zener Reference

Temperature coefficient of Zener diodes



- minimum $|TC|$ near 5 V ... 6 V: preferred reference Zeners 5.6 V, 6.2 V
- 6.2 V Zener ($+2$ mV/K) in series with a forward diode (-2 mV/K): compensated reference diode (1N821 family)

Feeding a Zener through a resistor

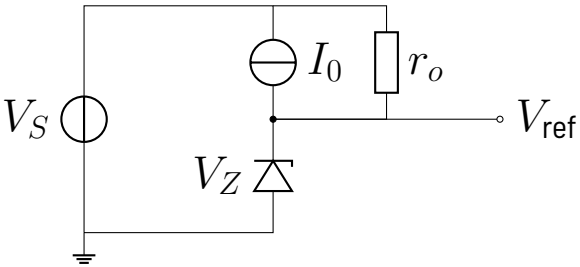


$$I_Z = I - I_L$$

- $R_V = \dfrac{V_S - V_Z}{I_Z + I_L}$, I_Z must stay above $I_{Z,\text{min}}$ for all V_S and I_L
- dynamic resistance $r_Z = \mathrm{d}V_Z/\mathrm{d}I_Z$: $5\,\Omega \ldots 50\,\Omega$
- line regulation $\dfrac{\Delta V_Z}{\Delta V_S} \approx \dfrac{r_Z}{R_V + r_Z}$, load regulation $\Delta V_Z \approx -r_Z\,\Delta I_L$

▶ Falstad: `examples/references/zener_resistor.txt`

Feeding a Zener from a current source



- the Zener current no longer depends on V_S
- only the output resistance r_o of a real current source remains: $\frac{\Delta V_Z}{\Delta V_S} \approx \frac{r_Z}{r_o}$
- e.g. transistor current source from chapter 3, $r_o = 100\,\text{k}\Omega$

▶ Falstad: `examples/references/zener_current_source.txt`

Exercise 4.1: Zener reference with series resistor

A Zener diode with $V_Z = 5.6\,\text{V}$ (at $5\,\text{mA}$) is fed from $V_S = 12\,\text{V}$ through $R_V = 1.2\,\text{k}\Omega$. Use the Falstad model of the Zener, $I_Z = 5\,\text{mA} \cdot e^{(V - V_Z)/V_T}$ with $V_T = 25.9\,\text{mV}$.

a) Calculate the Zener current and the dynamic resistance r_Z.
b) How much does V_{ref} change if V_S rises to $14\,\text{V}$?
c) A load $R_L = 2.2\,\text{k}\Omega$ is connected at $12\,\text{V}$. How much does V_{ref} drop?

Solution 4.1

a) $I_Z \approx (12\,\text{V} - 5.6\,\text{V})/1.2\,\text{k}\Omega = 5.33\,\text{mA}$, so $V_{\text{ref}} = V_Z + V_T \ln(5.33/5) = 5.602\,\text{V}$ and $r_Z = V_T/I_Z = 25.9\,\text{mV}/5.33\,\text{mA} = 4.9\,\Omega$.

b) At $14\,\text{V}$: $I_Z = 7.0\,\text{mA}$, $\Delta V_{\text{ref}} = V_T \ln(7.0/5.33) = 7.0\,\text{mV}$ (small-signal estimate with the mean $r_Z \approx 4.2\,\Omega$: $2\,\text{V} \cdot 4.2/1204 = 7.0\,\text{mV}$). Line regulation $3.5\,\text{mV/V}$, i.e. $0.06\,\%/\text{V}$.

c) The load takes $I_L = 5.6\,\text{V}/2.2\,\text{k}\Omega = 2.55\,\text{mA}$, the Zener keeps $I_Z = 5.33 - 2.55\,\text{mA} = 2.79\,\text{mA}$: $\Delta V_{\text{ref}} = V_T \ln(2.79/5.33) = -16.6\,\text{mV}$.

▶ Falstad: `examples/references/zener_resistor.txt`: the labels show `Vz12` $= 5.602\,\text{V}$, `Vz14` $= 5.609\,\text{V}$ ($+7.0\,\text{mV}$) and `Vz12L` $= 5.585\,\text{V}$ ($-16.6\,\text{mV}$).

Exercise 4.2: Zener with current-source feed

The Zener of the previous exercise is fed from a current source of 5 mA with an output resistance $r_o = 100\,\text{k}\Omega$. How much does V_{ref} change when V_S rises from 12 V to 14 V? Compare with the series resistor.

Solution 4.2

a) The current through r_o changes by $2\,\text{V}/100\,\text{k}\Omega = 20\,\mu\text{A}$, the Zener current from $5.064\,\text{mA}$ to $5.084\,\text{mA}$. $\Delta V_{\text{ref}} = V_T \ln(5.084/5.064) = 0.10\,\text{mV}$, 70 times less than with $R_V = 1.2\,\text{k}\Omega$ ($7.0\,\text{mV}$).

▶ Falstad: `examples/references/zener_current_source.txt`: at $12\,\text{V}$ `VzI` $= 5.600\,\text{V}$; moving the slider to $14\,\text{V}$ changes `VzR` by $7.0\,\text{mV}$ and `VzI` by only $0.1\,\text{mV}$.

4.4 The Bandgap Reference

Two voltages with opposite temperature coefficients

- base-emitter voltage: $V_{BE} = V_T \ln(I_C/I_S)$ with $V_T = kT/q = 25.7\,\text{mV}$ at $25\,^\circ\text{C}$
  - I_S rises strongly with T: V_{BE} falls by about $2\,\text{mV/K}$ (CTAT: complementary to absolute temperature)
- two transistors with current densities in the ratio N:

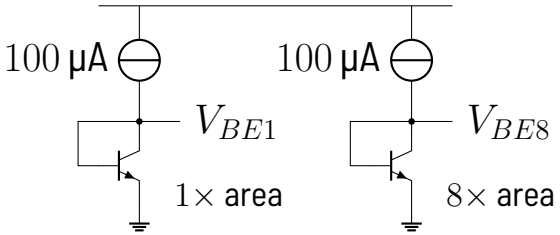
Delta-VBE

$$\Delta V_{BE} = V_{BE1} - V_{BE2} = V_T \ln N = \frac{kT}{q} \ln N$$

(PTAT: proportional to absolute temperature)

  - $k/q = 86.2\,\mu\text{V/K}$; $N = 8$: $\Delta V_{BE} = 53.5\,\text{mV}$ at $25\,^\circ\text{C}$, $+0.179\,\text{mV/K}$
  - I_S cancels: ΔV_{BE} depends only on T and N

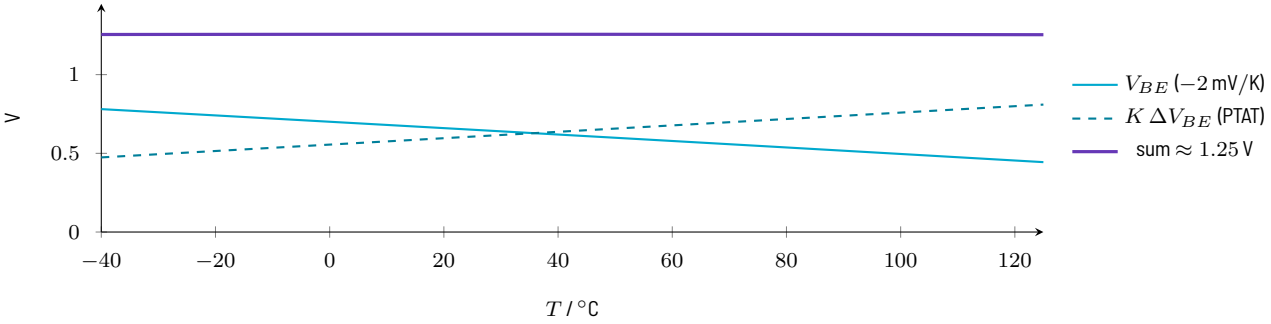
Delta-VBE in Falstad



- same current, 8 transistors in parallel: current density $1/8$
- Falstad has no temperature, but ΔV_{BE} at room temperature can be measured: $V_{BE1} = 0.536\,\text{V}$, $V_{BE1} - V_{BE8} = 53.8\,\text{mV}$
- current ratio 10: $\Delta V_{BE} = V_T \ln 10 = 59.6\,\text{mV}$ ("$60\,\text{mV}$ per decade")

▶ Falstad: `examples/references/delta_vbe.txt`

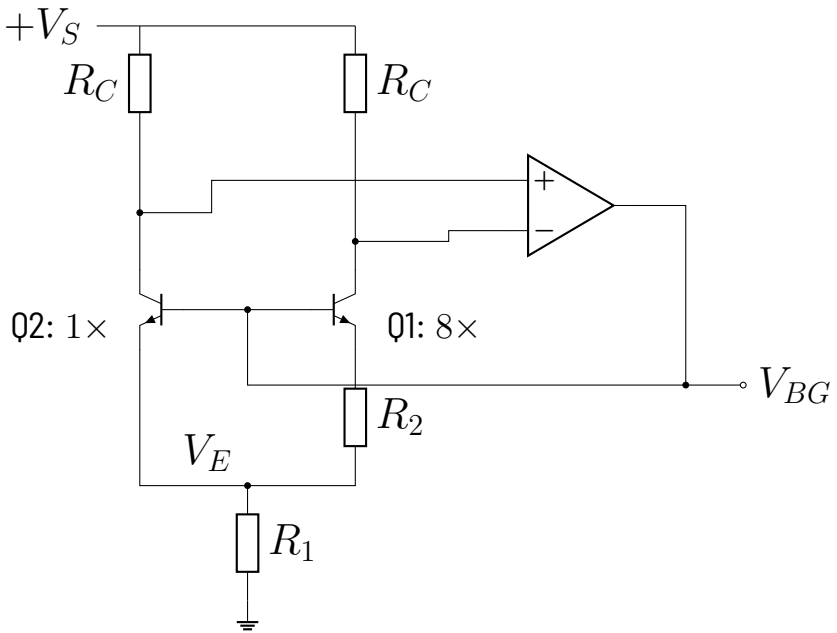
The bandgap principle



Bandgap reference

$V_{BG} = V_{BE} + K \cdot V_T \ln N \approx 1.25\,\text{V}$, K chosen so that $\mathrm{d}V_{BG}/\mathrm{d}T = 0$ at room temperature

The Brokaw cell



Brokaw cell: how it works

- the op-amp forces equal collector currents I (R_C equal, inputs at the collectors)
- current density ratio $N = 8$: $\Delta V_{BE} = V_T \ln 8$ appears across R_2
  - $I = V_T \ln 8 / R_2$ (PTAT current)
- R_1 carries both emitter currents: $V_E = 2IR_1 = 2\,\dfrac{R_1}{R_2}\,V_T \ln 8$
- output: $V_{BG} = V_{BE2} + 2\dfrac{R_1}{R_2}\,V_T \ln 8$ ($K = 2R_1/R_2$)
- Falstad, $R_2 = 1\,\text{k}\Omega$, $R_1 = 5.6\,\text{k}\Omega$: $I = 53.8\,\mu\text{A}$, $V_E = 0.602\,\text{V}$, $V_{BE2} = 0.520\,\text{V}$, $V_{BG} = 1.122\,\text{V}$

▶ Falstad: `examples/references/brokaw_cell.txt`

Exercise 4.3: Bandgap design

A Brokaw cell has an area ratio $N = 8$, $R_2 = 1\,\text{k}\Omega$ and $R_1 = 5.6\,\text{k}\Omega$. Use $V_T = 25.865\,\text{mV}$ (as Falstad).

a) Calculate ΔV_{BE}, the collector current I and V_E.
b) The Falstad transistor model gives $V_{BE2} = 0.520\,\text{V}$ at this current. Calculate V_{BG}.
c) Calculate the temperature coefficient of the PTAT part and compare it with $\text{d}V_{BE}/\text{d}T \approx -2\,\text{mV/K}$.

Solution 4.3

a) $\Delta V_{BE} = 25.865\,\text{mV} \cdot \ln 8 = 53.8\,\text{mV}$, $I = 53.8\,\text{mV}/1\,\text{k}\Omega = 53.8\,\mu\text{A}$, $V_E = 2 \cdot 53.8\,\mu\text{A} \cdot 5.6\,\text{k}\Omega = 0.602\,\text{V}$.

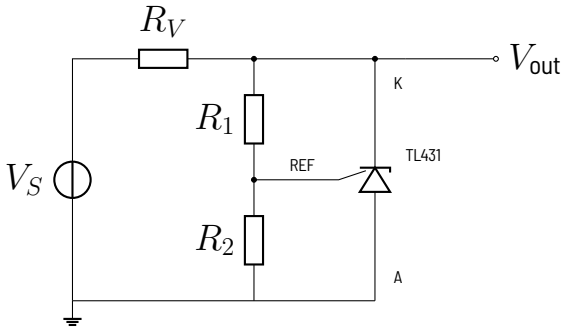
b) $V_{BG} = 0.520\,\text{V} + 0.602\,\text{V} = 1.122\,\text{V}$.

c) $\mathrm{d}V_E/\mathrm{d}T = 2\frac{R_1}{R_2}\frac{k}{q}\ln 8 = 11.2 \cdot 86.2\,\mu\text{V/K} \cdot 2.08 = +2.0\,\text{mV/K}$: it cancels the TC of V_{BE}.

▶ Falstad: `examples/references/brokaw_cell.txt`: VBG $= 1.122$ V, VE $= 0.602$ V; C1 and C2 are equal (4.47 V), the voltage across R_2 is 53.8 mV.

4.5 Shunt and Series References

The adjustable shunt reference TL431



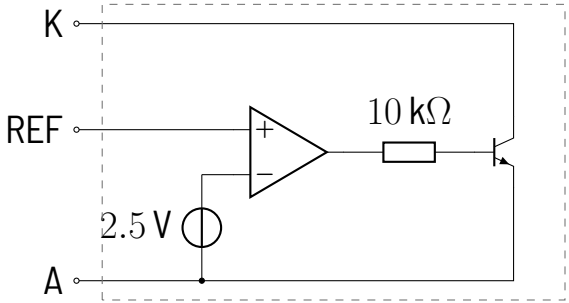
- the shunt element conducts until $V_{\text{REF}} = 2.5\,\text{V}$

TL431

$$V_{\text{out}} = V_{\text{REF}} \left(1 + \frac{R_1}{R_2}\right) + I_{\text{REF}}\, R_1, \qquad V_{\text{REF}} = 2.495\,\text{V},\ I_{\text{REF}} \approx 2\,\mu\text{A}$$

- cathode current between 1 mA and 100 mA

Inside the TL431 (behavioural model)



- error amplifier compares REF with the internal $2.5\,\text{V}$ bandgap
- V_{REF} too high $\Rightarrow$ transistor conducts more $\Rightarrow$ V_{out} falls
- $R_1 = 12\,\text{k}\Omega$, $R_2 = 20\,\text{k}\Omega$: $V_{\text{out}} = 2.5 \cdot 1.6 = 4.00\,\text{V}$

▶ Falstad: `examples/references/shunt_ref.txt`

Shunt or series reference?

- **shunt** (TL431, LM4040, LM385)
  - two terminals, works at any supply above V_{out}, can be used as a negative reference
  - the series resistor wastes current: R_V must deliver $I_{\text{min}} + I_{L,\text{max}}$ at the lowest supply
- **series** (REF50xx, ADR4xx, MAX6xxx)
  - three terminals, quiescent current only 0.1 mA … 1 mA
  - better for varying loads and battery operation, needs $V_{\text{in}} > V_{\text{out}} + V_{\text{dropout}}$
- both: decouple according to the data sheet (some are unstable with the wrong capacitor)

Exercise 4.4: TL431 set to 4.00 V

A TL431 ($V_{\text{REF}} = 2.495$ V, $I_{\text{REF}} = 2$ µA, $I_{K,\text{min}} = 1$ mA) shall provide 4.00 V from a supply of 6 V … 15 V.

a) Choose R_1 and R_2 from the E24 series.
b) How large is the error caused by I_{REF} and by $V_{\text{REF}} = 2.495$ V instead of 2.5 V?
c) Choose R_V such that a load of 1 mA can be supplied at 6 V.

Solution 4.4

a) $1 + R_1/R_2 = 1.6 \Rightarrow R_1/R_2 = 0.6$: $R_1 = 12\,\text{k}\Omega$, $R_2 = 20\,\text{k}\Omega$ (both E24).

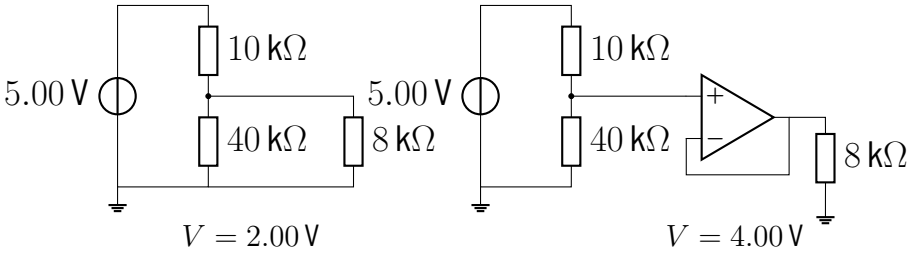
b) $I_{\text{REF}} R_1 = 2\,\mu\text{A} \cdot 12\,\text{k}\Omega = 24\,\text{mV}$ ($+0.6\,\%$); $2.495\,\text{V} \cdot 1.6 = 3.992\,\text{V}$ ($-0.2\,\%$). Together $V_{\text{out}} = 4.016\,\text{V}$: for an accurate $4.00\,\text{V}$ choose smaller divider resistors or trim.

c) At $6\,\text{V}$ R_V must carry $I_{K,\text{min}} + I_L + I_{\text{div}} = 1 + 1 + 0.125 = 2.125\,\text{mA}$: $R_V \leq (6\,\text{V} - 4\,\text{V})/2.125\,\text{mA} = 941\,\Omega$, choose $910\,\Omega$. At $15\,\text{V}$ the TL431 then carries $(15 - 4)/0.91 - 0.125 - 1 = 11\,\text{mA}$ ($44\,\text{mW}$): fine.

▶ Falstad: `examples/references/shunt_ref.txt` (ideal model with $2.5\,\text{V}$, $R_V = 1\,\text{k}\Omega$, no load): `VK` $= 4.00\,\text{V}$ at $12\,\text{V}$ and at $6\,\text{V}$, `REF` $= 2.50\,\text{V}$.

4.6 Buffering and Scaling a Reference

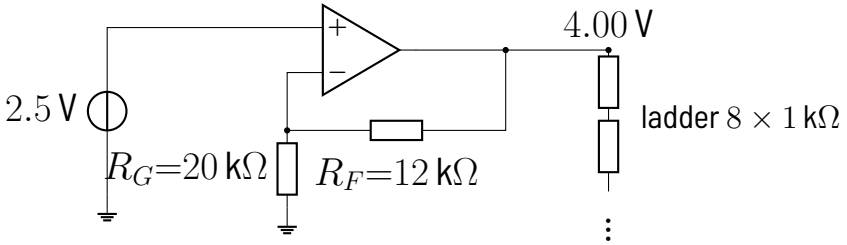
A divider is no reference



- divider $10\,\text{k}\Omega/40\,\text{k}\Omega$: 4.00 V open, but only 2.00 V with the $8\,\text{k}\Omega$ ladder
- source resistance $R_1 \parallel R_2 = 8\,\text{k}\Omega$ = load resistance
- follower: input current ≈ 0, output resistance ≈ 0: 4.00 V

▶ Falstad: `examples/references/ref_divider_loaded.txt`

Scaling with a non-inverting amplifier



- $G = 1 + R_F/R_G = 1 + 12/20 = 1.6$: $\quad 2.5\,\text{V} \cdot 1.6 = 4.00\,\text{V}$
- the op-amp delivers ladder current 0.5 mA plus 0.125 mA for $R_F + R_G$
- gain error from resistor tolerance: $\frac{\Delta G}{G} = \frac{G-1}{G}\left(\frac{\Delta R_F}{R_F} - \frac{\Delta R_G}{R_G}\right)$

▶ Falstad: `examples/references/ref_buffer_ladder.txt`

Design card: Reference for an ADC ladder

- full scale V_{FS} and LSB: LSB $= V_{\text{FS}}/2^N$ ($4.00\,\text{V}$, 3 bit: $0.5\,\text{V}$)
- ladder current $I = V_{\text{FS}}/(2^N R)$ ($8 \times 1\,\text{k}\Omega$: $0.5\,\text{mA}$)
- reference voltage V_R from a standard value, gain $G = V_{\text{FS}}/V_R$ (2.5 V, $G = 1.6$)
- non-inverting amplifier: $R_F/R_G = G - 1$, $R_F + R_G \gg$ ladder ($12\,\text{k}\Omega + 20\,\text{k}\Omega$)
- check: op-amp output current, $(G-1)/G\cdot$ resistor tolerance, $G \cdot V_{\text{offset}}$
- decouple the ladder top with $100\,\text{nF}$ ... $1000\,\text{nF}$ close to the comparators

Exercise 4.5: Reference for the 3-bit flash converter

The flash converter of the capstone needs $4.00\,\text{V}$ at the top of a ladder of $8 \times 1\,\text{k}\Omega$. A $2.5\,\text{V}$ reference is available.

a) Calculate the ladder current and the tap voltages.
b) Design a non-inverting amplifier with E24 resistors.
c) Why can the $4.00\,\text{V}$ not be taken from a divider $10\,\text{k}\Omega/40\,\text{k}\Omega$ at a $5\,\text{V}$ reference?
d) How large is the worst-case error of V_{ref} with $1\,\%$ resistors?

Solution 4.5

a) $I = 4\,\text{V}/8\,\text{k}\Omega = 0.5\,\text{mA}$; taps every $0.5\,\text{V}$: $T_1 = 0.5\,\text{V}$, ..., $T_4 = 2.0\,\text{V}$, ..., $T_7 = 3.5\,\text{V}$.

b) $G = 4/2.5 = 1.6$, $R_F/R_G = 0.6$: $R_F = 12\,\text{k}\Omega$, $R_G = 20\,\text{k}\Omega$.

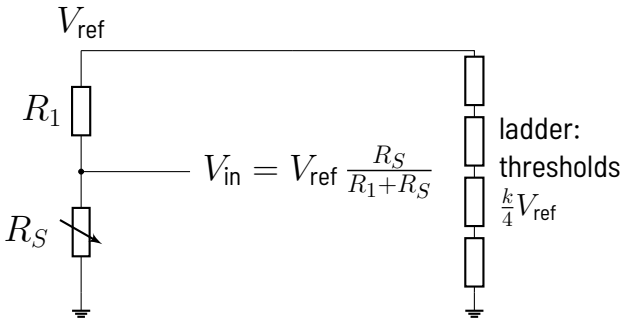
c) The divider has a source resistance of $10\,\text{k}\Omega \parallel 40\,\text{k}\Omega = 8\,\text{k}\Omega$, equal to the ladder. The loaded voltage is $5 \cdot \frac{40 \parallel 8}{10+40 \parallel 8} = 5 \cdot \frac{6.67}{16.67} = 2.00\,\text{V}$.

d) $\Delta G/G = 0.375 \cdot 2 \cdot 1\,\% = 0.75\,\%$, i.e. $\pm 30\,\text{mV}$; for the 3-bit converter (LSB $0.5\,\text{V}$) irrelevant, for a 12-bit converter 30 LSB.

▶ Falstad: `examples/references/ref_buffer_ladder.txt`: `VREF4` $= 4.00\,\text{V}$, `T4` $= 2.00\,\text{V}$, `T1` $= 0.50\,\text{V}$, ladder current $0.5\,\text{mA}$;
▶ Falstad: `examples/references/ref_divider_loaded.txt`: `Vopen` $= 4.00\,\text{V}$, `Vload` $= 2.00\,\text{V}$, `Vbuf` $= 4.00\,\text{V}$.

4.7 Ratiometric Measurement

The reference cancels



Ratiometric

$$\text{code} = 2^N \frac{V_{\text{in}}}{V_{\text{ref}}} = 2^N \frac{R_S}{R_1 + R_S}$$ independent of V_{ref}

Ratiometric versus absolute

- Falstad: two 2-bit flash converters (3 comparators), sensor divider $1\,\mathrm{k\Omega}/2.7\,\mathrm{k\Omega}$: $V_{\mathrm{in}} = 0.730\,V_{\mathrm{exc}}$
- left: sensor and ladder from V_{ref}: code 2 for $V_{\mathrm{ref}} = 3.6, 4.0, 4.4\,\mathrm{V}$
- right: sensor from a fixed $4.00\,\mathrm{V}$, ladder from V_{ref}
  - $V_{\mathrm{ref}} = 4.0\,\mathrm{V}$: $V_{\mathrm{in}} = 2.92\,\mathrm{V} < 3.0\,\mathrm{V}$: code 2
  - $V_{\mathrm{ref}} = 3.6\,\mathrm{V}$: threshold $\frac{3}{4}V_{\mathrm{ref}} = 2.7\,\mathrm{V}$: code 3 (wrong)
- ratiometric works only if both see the **same** voltage at the **same** time
  - a common connection with voltage drops, or noise on one path only, breaks it

▶ Falstad: `examples/references/ratiometric.txt`

4.8 Error Budget

Error budget of the 4.00 V reference

contribution	data	error
initial accuracy	$\pm 0.2\,\%$	$0.20\,\%$
temperature	25 ppm/K, ± 25 K	$0.06\,\%$
gain resistors	$2 \times 0.1\,\%$, factor 0.375	$0.08\,\%$
op-amp offset	1 mV $\times$ 1.6	$0.04\,\%$
long-term drift	50 ppm / 1000 h	$0.01\,\%$
worst case (sum)		$0.39\,\%$
RSS		$0.23\,\%$

- compare: 3 bit, LSB $= 12.5\,\%$ of full scale: no problem at all
- 12 bit, LSB $= 0.024\,\%$: 16 LSB worst case $\Rightarrow$ calibrate or measure ratiometrically

Exercise 4.6: Ratiometric measurement and error budget

A sensor divider $R_1 = 1\,\text{k}\Omega$, $R_S = 2.7\,\text{k}\Omega$ is read by a 2-bit flash converter with a ladder of $4 \times 1\,\text{k}\Omega$.

a) The divider and the ladder are fed from the same V_{ref}. Which code results for $V_{\text{ref}} = 4.0\,\text{V}$ and $3.6\,\text{V}$?

b) The divider is fed from a fixed $4.00\,\text{V}$, the ladder from V_{ref}. Repeat a).

c) A 12-bit converter uses a reference with $\pm 0.2\,\%$ initial accuracy and $25\,\text{ppm/K}$. How many LSB can the result be off at $50\,^\circ\text{C}$ after calibration at $25\,^\circ\text{C}$?

Solution 4.6

a) $V_{\text{in}}/V_{\text{ref}} = 2.7/3.7 = 0.730$; the thresholds are at 0.25, 0.5 and $0.75\,V_{\text{ref}}$. $0.5 < 0.730 < 0.75$: code 2 for every V_{ref}.

b) $V_{\text{in}} = 2.92\,\text{V}$. $V_{\text{ref}} = 4.0\,\text{V}$: thresholds 1, 2, 3 V: code 2. $V_{\text{ref}} = 3.6\,\text{V}$: thresholds 0.9, 1.8, $2.7\,\text{V}$: code 3. A $10\,\%$ reference error changes the result by one code.

c) After calibration only the drift remains: $25\,\text{ppm/K} \cdot 25\,\text{K} = 625\,\text{ppm} = 0.000625 \cdot 4096 = 2.6$ LSB. Without calibration add $0.002 \cdot 4096 = 8.2$ LSB.

▶ Falstad: `examples/references/ratiometric.txt`: with the `VREF` slider at 4.0 V both converters show D = E = code 2 (`VIN1` $= 2.92$ V); at 3.6 V the right converter switches `E3` on (code 3), the left stays at code 2.